

Python For Computational Problem Solving- UE26CS151A

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Leet Code Quick Guide

What is LeetCode?

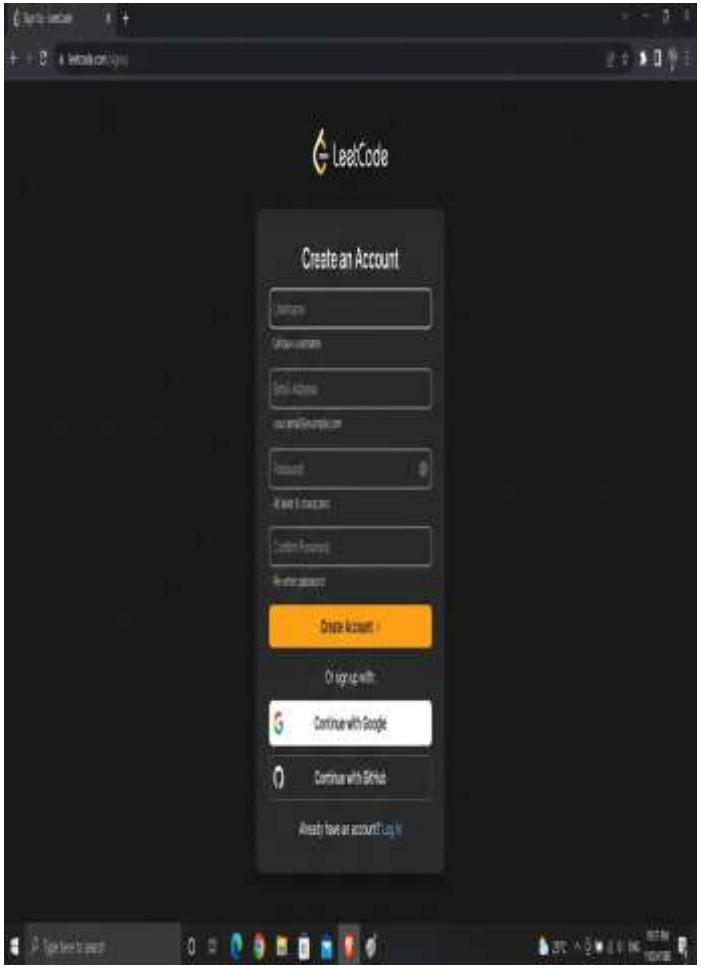
Platform Overview

- **Industry Benchmark:** Global platform used by software engineers and students to practice coding and data structures.
- **3,000+ Problems:** Curated challenges across arrays, strings, trees, dynamic programming, and math.
- **Automated Evaluation:** Instant testing against hidden test cases with execution time and memory feedback.
- **Multi-Language Support:** Supports Python 3, C++, Java, and more (Python 3 is our primary language).

Why Practice on LeetCode?

- **Interview Preparation:** Primary technical screening benchmark for top companies (Google, Microsoft, Amazon).
- **Active Problem Solving:** Moves learning from passive syntax reading to active algorithmic thinking.
- **Real-Time Feedback:** Instantly flags logical bugs, edge cases, and runtime inefficiencies.
- **Structured Curricula:** Offers high-yield tracks like LeetCode 75 and Top Interview 150.

How to Create a LeetCode Account



Quick Sign-up Steps

- **Step 1: Visit Portal:** Go to leetcode.com and click 'Sign up'.
- **Step 2: Enter Details:** Provide a unique Username, Email, and secure Password.

Note: Username should be your PRN/SRN only

- **Step 3: Pick Good Handle:** Use a professional handle (e.g. name_pes) for resumes.
- **Step 4: SSO Option:** Optionally register in 1-click using Google or GitHub.
- **Step 5: Verify Email:** Check your inbox and confirm email to activate account.

"Kindly note and securely save the credentials for the future Use".

Profile Section, Daily Streaks & Badges

Profile & Settings

- **Set Python 3 Default:** Choose Python 3 in the code editor once; it stays default forever.
- **Personal Bio & Links:** Add your PES University details, GitHub profile, and LinkedIn.
- **Problem Statistics:** Tracks solved counts broken down by Easy, Medium, and Hard.
- **Activity Heatmap:** Green contribution calendar showing your active daily solving history.

Streaks, Badges & Coins

- **Daily Challenge Streak:** 1 new curated problem posted daily at 05:30 AM IST. Solve it to build streak.
- **Achievement Badges:** Earn monthly badges and study-plan badges showcased on your profile.
- **LeetCode Coins:** Earn coins for daily check-ins; redeem for interview guides and profile perks.
- **GitHub Auto-Sync:** Use the 'LeetHub' Chrome extension to auto-push solved solutions to your GitHub!

Key Features of the LeetCode Platform

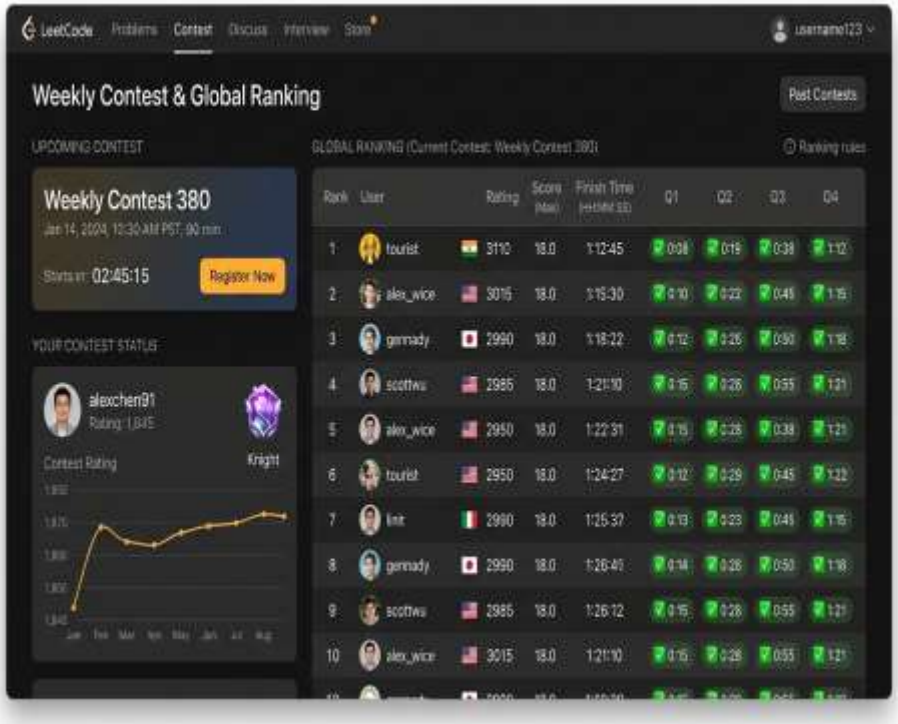
Problem Navigation Features

- **Difficulty Tiers:** Easy (syntax & basic structures), Medium (interview core), Hard (complex DP & graphs).
- **Topic Filtering:** Filter questions by topic: Arrays, Two Pointers, Hash Table, Stack, Trees, DP.
- **Curated Study Plans:** Follow structured paths: LeetCode 75, Top 150, and 30 Days of Pandas.
- **Company Problem Tags:** Filter questions commonly asked by specific tech companies.

Developer Environment & IDE

- **In-Browser IDE:** Dark mode code editor with syntax highlighting, line numbers, and auto-indentation.
- **Custom Testcase Console:** Run edge cases locally in the Testcase tab before official submission.
- **PEP 8 Compliance:** Default 4-space tab indentation for clean, readable Python code.
- **IntelliSense Autocomplete:** Code suggestions and parameter hints inside the browser editor.

Contests, Rating System & Global Rankings



Contest & Rating Rules

- **Weekly & Biweekly Rounds:** Live 90-min contests (Sunday 8:00 AM IST & Saturday 8:00 PM IST).
- **4-Problem Scoring:** Q1 Easy (3 pts), Q2-Q3 Medium (4-5 pts), Q4 Hard (6-7 pts). Total ~18-19 pts.
- **5-Minute Penalty:** Each failed submission adds 5 mins to finish time (only for solved problems).
- **Knight & Guardian:** Global Elo rating awards badges: Knight (Top 25%) and Guardian (Top 1.5%).
- **Virtual Contest Mode:** Simulate past contests anytime with real-time countdown timer and leaderboard.

Discuss, Interview Prep & Study Plans

Interview Prep & Community

- **Company Interview Experiences:** Candidates share real-world questions, interview rounds, and offers from Google, Amazon, etc.
- **Official Editorial:** Staff-written mathematical proofs, multiple approaches, and Big-O derivations.
- **Solutions / Discuss Tab:** Explore top-voted community code in Python 3 to learn idiomatic one-liners and patterns.
- **Custom Problem Lists:** Bookmark questions into custom revision lists (e.g. 'PES Lab', 'Must-Do DP').

Explore Cards & Career Roadmaps

- **Explore Learning Tracks:** Interactive guided modules for Arrays 101, Binary Search, Trees, and Recursion.
- **High-Yield Study Plans:** Structured roadmaps: LeetCode 75, Top Interview 150, and 30 Days of Pandas.
- **Company Problem Frequency:** Sort problems by how frequently top employers asked them in recent interviews.
- **Mock Interview Simulator:** Timed 45-60 min assessments replicating actual company phone screens.

How to Solve a Question: 4-Step Method

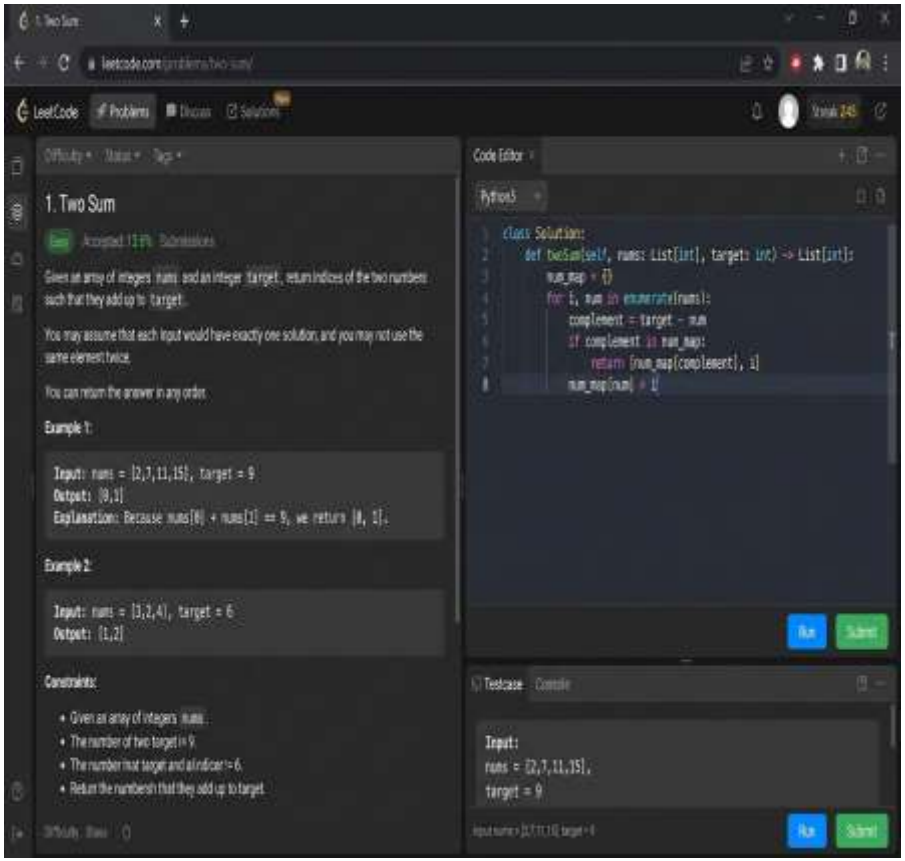
Step 1 & Step 2: Understand & Plan

- **Step 1: Read & Check Bounds:** Read problem carefully. Inspect constraints ($N \leq 10^5$ means avoid $O(N^2)$ loops).
- **Check Edge Cases:** Ask: Can array be empty? Negative numbers? Duplicate elements? Single element?
- **Step 2: Choose Approach:** Think of naive brute-force first, then use optimal data structures (Hash Map, Set).
- **Determine Big-O:** Make sure your time complexity executes in < 1 second ($< 10^7$ operations).

Step 3 & Step 4: Code & Verify

- **Step 3: Write Clean Code:** Write solution inside `class Solution:`. Use `return`, NOT `print()`!
- **Step 4: Run Code (Sandbox):** Click 'Run Code' to test against sample cases. Add custom cases in Testcase tab.
- **Submit Solution:** Click green 'Submit' button to run across 50–200+ hidden test cases.
- **Review & Learn:** If Accepted, check runtime percentile and read Editorial for better tricks.

LeetCode Problem Solving Workspace



Workspace Breakdown

- **1. Description Panel (Left):** Contains problem narrative, sample input/outputs, and numerical constraints.
- **2. Language Selector:** Dropdown at top of editor; ensure 'Python 3' is selected.
- **3. Code Editor (Right):** Write your solution inside the given class and method template.
- **4. Run Code (Blue):** Tests your code against sample inputs without penalty.
- **5. Submit (Green):** Evaluates code against hidden test cases for final verdict.



Thank You

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